Polyphase magnetization of the Big Spruce Complex, Northwest Territories[1]

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The Big Spruce alkaline complex has yielded a Rb–Sr isochron of 2111 ± 40 m.y. The rocks are unmetamorphosed, but the local geochronological evidence indicates that they have had a complicated thermal history. The complex has four magnetizations, which can be separated by a two-stage demagnetization procedure. The **D** magnetization (19 sites; 022, +88; α_{95} =8°; pole 67° N 113° W) is considered to have been acquired at the time of emplacement because it generally has the highest blocking temperatures (500 to 650 °C). The **X**, **Y**, and **Z** magnetizations are believed to be later overprints, as they generally have lower blocking temperatures and have directions that are comparable to those found in younger rocks in the interval of approximately 2000 to 1400 m.y. Most of the overprints seem to have been acquired at times of mild reheating during this interval.

Le complexe alcalin de Big Spruce a fourni une isochrone Rb–Sr de 2111 ± 40 Ma. Les roches ne sont pas métamorphisées, mais présentent des indices géochronologiques locaux d'une histoire thermale compliquée. Le complexe a quatre magnétisations, que l'on peut distinguer par une démagnétisation en deux stades. La magnétisation **D** (19 sites; 022, + 88; α_{95} = 8°; pôle à 67° N 113° W) a dû être acquise à l'époque de la mise en place; elle a en effet les plus hautes températures de blocage (500 à 650 °C). Les magnétisations **X**, **Y**, et **Z** sont probablement ultérieures: elles ont généralement des températures de blocage inférieures et ont des directions comparables à celles que l'on trouve dans des roches plus jeunes, dans l'intervalle de 2000 à 1400 Ma environ. La plupart des surimpressions magnétiques semblent avoir été acquises durant cet intervalle, lors de périodes de légers réchauffements du complexe.

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Introduction

Polyphase magnetization is the phenomenon whereby a rock unit may contain several different magnetizations, each corresponding to the geomagnetic field at various times during the history of the rock unit. These magnetizations may be present in different oxide phases, or in different grain-size ranges of the same oxide phase. They can be separated by chemical leaching, which progressively removes the finer grain-size fractions and the magnetizations they carry; by alternating field (AF) demagnetization, which progressively removes the magnetizations of lower remanent coercive force; or by thermal demagnetization, which progressively removes the magnetizations with lower blocking temperatures. The phenomenon of polyphase magnetization is analogous to polyphase deformation in metamorphic rocks, which is indicative of a complex metamorphic history, and to the occurrence of a spread of apparent dates obtained in geochronological studies, which is indicative of a complex thermal history.

One of the simplest situations in paleomagnetism is that in which there is a primary magnetization, on which a second magnetization due to the present field is imposed. Such secondary magnetizations can be readily removed by partial AF or thermal demagnetization. The next most complex case is that in which magnetizations have been acquired at some more remote time, usually during deep burial and uplift; Permian overprints in Devonian and Silurian red beds (Chamalaun 1964; Irving and Opdyke 1965), and old overprints in Precambrian rocks (McMurry *et al.* 1973; Buchan and Dunlop 1973; Irving *et al.* 1974) have been documented. However, the magnetization of rocks that have been subjected to a series of episodes of reheating or chemical change over an interval during which the field has greatly changed, is much more complex. The primary magnetization may have been partially or completely replaced by two or more secondary magnetizations, and the identification of these magnetizations, and the determination of their order in time, are the most important technical problems in Precambrian paleomagnetism. Recently, Roy and Park (1974) have shown that certain Carboniferous red beds deposited during an interval when the geomagnetic field was reversing frequently contained three magnetizations (two reversed and one normal),

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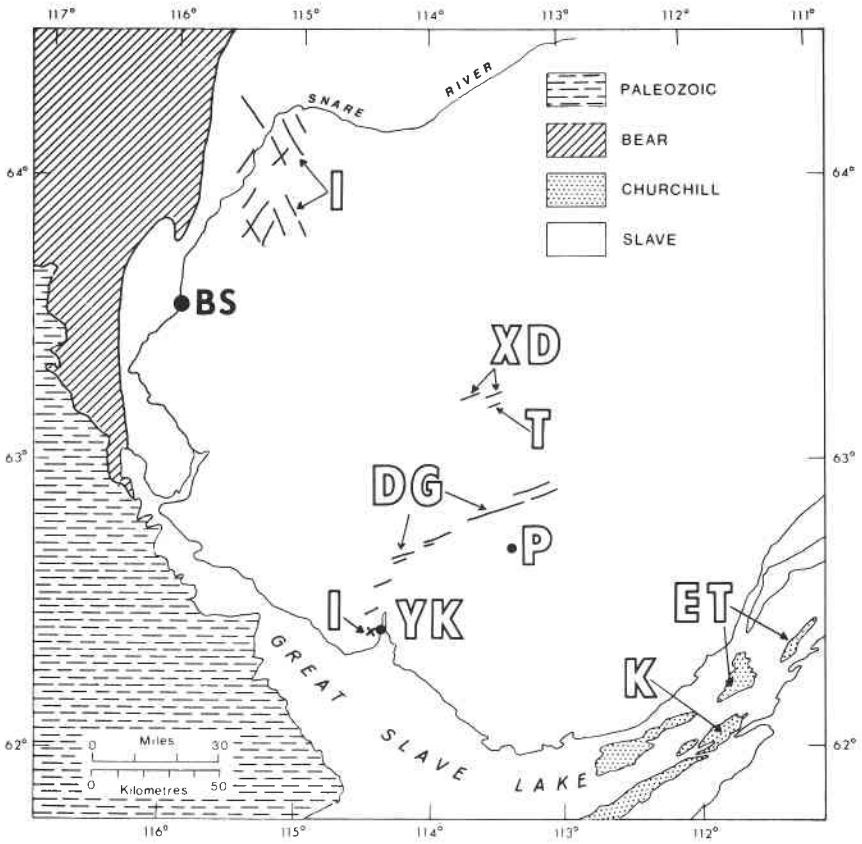


FIG. 1. Sketch-map showing general location of the Big Spruce Complex (BS). Other rock units from which paleomagnetic results are available are also shown: Dogrib (DG) Indin (I) and 'X' (XD) dikes (McGlynn and Irving 1975); Thistlewaite diabase dike (T) and Pensive Lake diabase sheet (P) (McGlynn and Irving 1975); Et-then Group red beds (Irving *et al.* 1972); Kahochella Group red-beds (McMurry *et al.* 1973). YK is the town of Yellowknife.

and Park (1975) has recognized four magnetizations in the metamorphic rocks near Flin Flon. In this paper we describe an instance of polyphase magnetization in unmetamorphosed igneous rocks.

Geology

The Big Spruce Complex occurs about 140 km northwest of Yellowknife (Fig. 1). It intrudes Archean granitic and supracrustal rocks of the Slave Structural Province. Martineau and Lambert (1974) have shown that the northeastern part consists of silica-saturated layered diorites and gabbros intruded by magnetite-rich pyroxenites and a syenite ring dike (Fig. 2). The rest of the complex comprises silica-saturated and oversaturated syenites, and undersaturated foyaites and ijolites, which we refer to collectively as 'syenites'. There are local swarms of thin basic dikes and a carbonatite plug. Fenites are well developed about the margins of the silica-undersaturated rocks, and gas explosion vents with related breccias also occur near contacts. The intrusion is cut by a small fault, and there is some indication from the regional geology that it lies in a major fault zone. The rocks commonly show internal fracturing, but there is no evidence of thermal metamorphism.

Martineau and Lambert (1974) obtained an Rb–Sr whole-rock isochron from the complex yielding an age of 2111 $\pm$ 40 m.y. ($\lambda = 1.39 \times 10^{-11}$ years). They also obtained an isochron of 1925 $\pm$ 40 m.y. from a nearby small biotite pyroxenite complex. Martineau and Lambert reported K–Ar mineral ages of 1800 m.y., and a biotite collected by W. R. A. Baragar yielded a

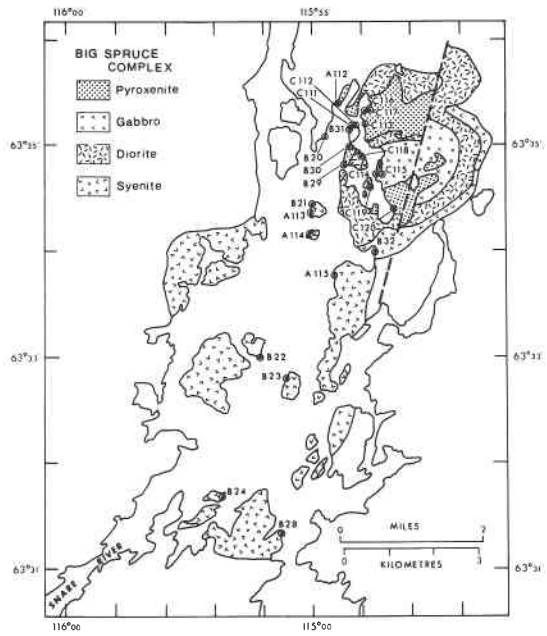


FIG. 2. Sketch-map of Big Spruce Complex, showing sampling localities. The base map is from Martineau and Lambert (1974). Collections were made during three field seasons: the A sites in 1971, the B sites in 1972, and the C sites in 1974. The numbers are field excession numbers and have no geological significance. Big Spruce Lake is a hydro-electric development on the Snare River and provides power to Yellowknife. The level has recently been raised so the Complex is now almost entirely under water.

K–Ar age of 1785 m.y. (Leech *et al.* 1963). The granites into which the complex is intruded yielded Rb–Sr isochron ages in the range of 2400–2600 m.y. In this paper, we have accepted 2111 ± 40 m.y. as the age of intrusion of the complex.

Procedures

Oriented cores (111 altogether) were collected from 22 sites and their distribution is shown on Fig. 2. About five cores were collected at each site, and two samples were sliced from each core. Orientations were measured by sun compass. Contact samples of Yellowknife Supergroup sedimentary rocks were collected at site A112 (the directions were close to the present field and became scattered in alternating fields of 100 Oe and yielded no useful information). Contact rocks were sought elsewhere, but not found. Magnetizations were measured using a Princeton Applied Research Spinner magnetometer (SM1) and Schonstedt spinner magnetometers (SSM1 and DSM1). Alternating field (AF peak amplitudes quoted) and thermal demagnetization was carried out using the apparatus described by Roy *et al.* (1969; 1971; 1973). A three-axis tumbler was used during AF demagnetization. Both thermal and AF demagnetization were carried out in fields of less than 2 gammas.

All specimens contain a large, magnetically soft component, which can be readily removed by alternating fields of about 100 Oe, and which commonly reduces the remanence to between 10% and 1% of its original value. The magnetization remaining is itself composite, but it cannot be resolved into its constituent magnetizations by further AF demagnetization. The technique that we have found most useful in analyzing these polyphase magnetizations is a two-stage demagnetization procedure (Park 1975). The large soft magnetizations were first removed by AF demagnetization, and then thermal demagnetization was used. Presumably, the reason why this two-stage demagnetization technique is successful while the application of AF demagnetization alone is not is that, in general, the remanent coercive force spectra of the constituent magnetizations overlap, but their blocking temperature spectra do not. The magnetization remaining after initial AF demagnetization is less than 10%, and sometimes less than 1%, of the original magnetization. Furthermore, its constituent magnetizations may comprise only a few parts of one per cent of the original magnetization, so that the acquisition of spurious magnetizations during demagnetization, due say to departures from perfect field cancellation, could lead to serious distortion of the results. Great care has been exercised to ensure that spurious magnetizations have been minimized and are random in direction. In any one thermal demagnetization run, the specimens are stored randomly in the furnace; they have been demagnetized in many different batches over a period of three years, using three different magnetometers. Between demagnetization and measurement, specimens have been stored in random orientations in mumetal shields, inside which the field is less than 5 gammas.

Resolution of X, Y, Z and D Magnetizations

The basic rocks from the north of the complex are very magnetic (Table 1). Their remanent magnetizations are complex and results from

TABLE 1. Results from each collecting locality

Loc.	Type	N	M	$D,I(R)$	$\delta°$	Treatment
A112	**D**	6	5	251,+75(5.94)	8	100 Oe
A113	**D**	5	666	013,+69(4.93)	9	200 Oe
A115	**Y**	5	37	153,+57(4.76)	18	100 and 200 Oe
B020	**Y**	3	5	131,+38(2.91)	14	500 °C after 300 Oe
B021	**D**	6	949	121,+82(5.91)	10	100 and 200 Oe
B022	**D**	4	179	265,+72(3.89)	13	ditto
B023	**D**	3	1329	011,+85(2.97)	7	ditto
B024	**D**	4	23	216,+73(3.92)	11	nrm and 200 Oe
B028	**D**	4	29	240,+85(3.99)	5	nrm and 100 Oe
B029	**D**	5	7	118,+80(4.82)	15	100 and 200 Oe
B030	**D**	4	34098	115,+78(3.93)	12	200 and 300 Oe
B031	**D**	5	29738	056,+71(4.93)	9	ditto
B032	**D**	2	11	248,+83(2.00)	—	300 Oe
C111	—	5	38510	193,+36(4.89)	12	400 Oe
C111	**X**	5	—	192,+28(4.80)	16	400 Oe—625 °C (V)
C111	**D**	5	—	168,+64(4.81)	16	650 °C after 400 Oe
C113	—	5	55974	263,+22(4.18)	33	400 Oe
C113	**Z**	4	—	278,−23(3.56)	27	400 Oe—600 °C (V)
C113	**D**	4	—	288,+65(3.83)	17	550 °C after 400 Oe
C114	—	5	1513	328,−25(4.76)	18	400 Oe
C114	**Y**	5	—	312,−43(4.86)	14	400 Oe—620 °C (V)
C114	**D**	5	—	037,+58(4.75)	18	620 °C after 400 Oe
C115	—	5	2577	215,+27(4.96)	7	nrm
C115	**X**	4	—	227,+23(3.87)	15	nrm—100 Oe (V)
C115	—	5	—	320,−19(4.94)	9	400 Oe
C115	**Y**	7	—	321,−41(4.87)	13	400 Oe—650 °C (V)
C115	**D**	5	—	321,+62(4.65)	22	620 °C after 400 Oe
C116	—	5	3866	297,+18(4.99)	4	300 Oe
C116	**Z**	5	—	292,+02(4.75)	18	300 Oe—620 °C (V)
C116	**D**	5	—	317,+64(4.65)	22	620 °C after 300 Oe
C117	—	4	3202	298,+11(3.98)	6	200 Oe
C117	**Z**	4	—	300,+04(3.97)	7	200 Oe—600 °C (V)
C117	**D**	4	—	133,+74(3.83)	17	650 °C after 200 Oe
C118	—	3	5	303,+70(2.87)	17	200 Oe
C118	**Z**	3	—	292,+26(2.62)	29	600 °C after 200 Oe
C119	**D**	4	13616	114,+76(3.97)	7	200 Oe
C119	**D**	5	—	108,+75(4.70)	20	550 °C after 200 Oe
C119	**D**	5	—	303,+84(4.54)	25	650 °C after 200 Oe
C120	**Y**	5	68	130,+48(4.87)	13	nrm*
C120	**D**	5	—	016,+84(4.69)	20	200 Oe
C120	**D**	5	—	078,+61(4.83)	15	620 °C after 200 Oe

In the second column, the magnetization type is given using the symbols **D**, **X**, **Y**, and **Z** as explained in the text (Fig. 12). N is the number of oriented cores. M is the intensity of nrm in microgauss (cg/cm^3 $\times$ 10^{-6}). D,I are the declination and inclination of the mean direction, and R is the resultant. $\delta°$ is Wilson's standard deviation (Wilson 1959). (V) is the vector analysis of the two treatments indicated. Results from localities with their numbers in italics have been used in the analysis of Table 3.

*The best estimate of **Y** is the nrm; subtraction of removed magnetizations produces no better estimate of **Y** because the ratio of nrm to the magnetization remaining after AF cleaning is very large.

three sites are described. The intensities of magnetization of the syenite are lower and their remanence simpler.

Pyroxenite from Site C111

The results of AF demagnetization of a specimen from pyroxenite at site C111 are shown in Fig. 3. Its direction is little changed in fields up to 600 Oe, and in higher fields the directions scatter. The intensity falls very rapidly, being less than 0.2% of the original value (the natural remanent magnetization nrm) in fields above 400 Oe. Over 99.8% of the nrm consists of a relatively soft magnetization that is directed to the SSW with low positive inclination. Magnetizations with this general direction are referred to as **X** magnetization. The **X** magnetization diverges from the present field by a large angle and therefore cannot be a viscous magnetization of recent origin.

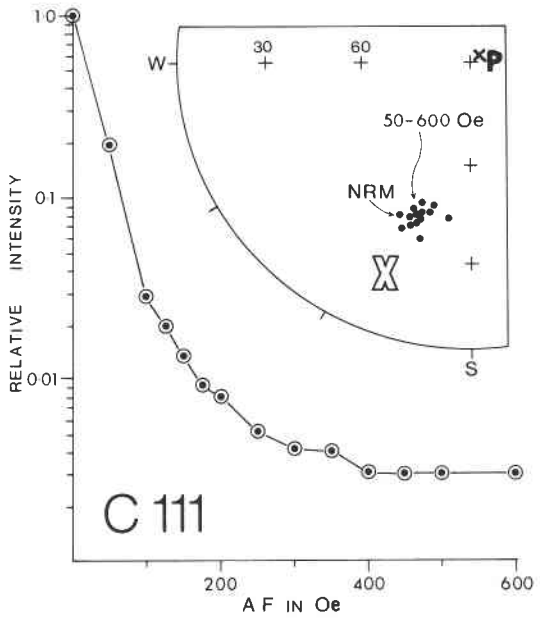


FIG. 3. Alternating field demagnetization of pyroxenite from site C111. The intensity is plotted on a log scale. The corresponding direction changes are shown as dots representing downward inclinations. The nrm intensity is 59 512 μG. P is the direction of the present field.

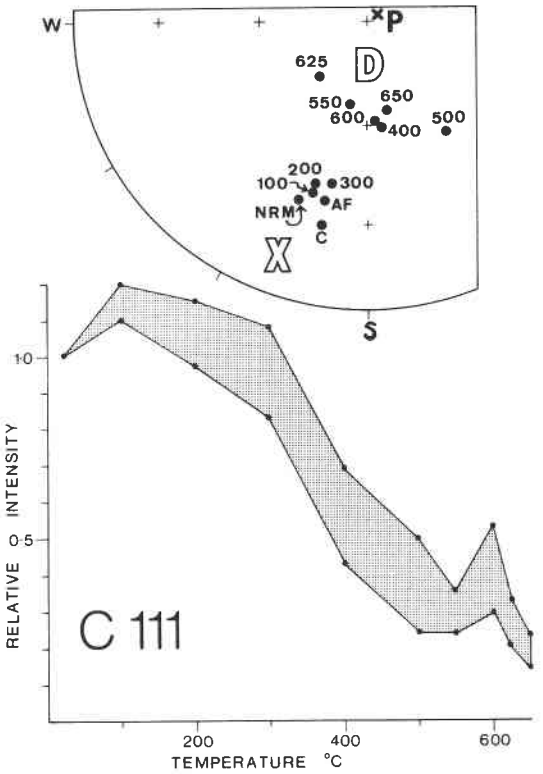


FIG. 4. Thermal demagnetization following AF demagnetization of pyroxenite from site C111. Mean directions for 5 core samples are given above; all have downward inclination. The point labelled NRM is the mean direction of nrm. Results have been obtained by thermal demagnetization of specimens that have first been partially demagnetized in 400 Oe. AF is their mean direction after treatment in AF of 400 Oe. The mean directions of 5 cores obtained after thermal demagnetization are labelled with their corresponding temperatures in °C. C is the direction of the X magnetization obtained by correcting for the effect of the D magnetization. The areas of the sterogram generally occupied by the X and D magnetizations are marked. The decay of intensity is shown below by the envelope defined by the standard error of 5 thermal demagnetization curves.

In Fig. 4, the resolution of the magnetization remaining after AF demagnetization in 400 Oe is shown. The average directions of all specimens from site C111 are shown. Thermal demagnetization up to 300 °C does not change the direction, but at 400 °C the inclinations increase abruptly and then remain essentially constant up to 650 °C. Magnetizations with steeply inclined downward directions are referred to as the **D** magnetizations. The corresponding intensity changes are given in the lower part of Fig. 4. The intensity falls by a factor of 3, and is less than 0.1% of the nrm after thermal demagnetization between 500 and 650 °C. Thus, that part of the **X** magnetization with remanent coercivities exceeding 400 Oe has blocking temperatures of less than 400 °C. Most of the **D** magnetization has blocking temperatures in excess of 650 °C. There may be some blocking temperatures in the range 500 to 650 °C, but the noise inherent in this extreme demagnetization process (as shown by the standard errors in the decay curves) do not allow this to be definitely determined. The directions of the **D** and **X** magnetizations at C111 are listed in Table 1. The **D** magnetization is the magnetization remaining after AF treatment followed by thermal treatment, and in Table 1 we have chosen the results at the 650 °C step as the best estimate of **D**, because at that step the scatter among specimens was the least. The **X** magnetization was obtained by subtracting vectorially the magnetization after AF demagnetization (labelled AF in Fig. 4) from the **D** magnetization. An example of results from a single specimen from C111 is given in Table 2. The vector subtractions were performed for all specimens for all thermal demagnetization steps from 500 to 650 °C and the value chosen (the

TABLE 2. Resolution of Polyphase Magnetizations

	M	*D,I*	Type
Site C111			
400 Oe	1104	208, +44	—
600 °C	195	226, +61	**D**
625 °C	234	206, +66	**D**
650 °C	140	163, +62	**D**
400 Oe—600 °C (V)	923	206, +40	**X**
400 Oe—625 °C (V)	891	208, +38	**X**
400 Oe—650 °C (V)	987	212, +41	**X**
Site C115			
400 Oe	95	314, −22	—
600 °C	100	333, +83	**D**
625 °C	64	336, +64	**D**
650 °C	55	272, +66	**D**
400 Oe—600 °C (V)	155	311, −60	**Y**
400 Oe—625 °C (V)	112	304, −56	**Y**
400 Oe—650 °C (V)	113	326, −50	**Y**
Site C117			
200 Oe	1090	297, +10	—
600 °C	214	307, +38	—
625 °C	89	358, +55	—
650 °C	90	021, +76	**D**
200 Oe—600 °C (V)	910	295, +04	**Z**
200 Oe—625 °C (V)	1056	295, +06	**Z**
200 Oe—650 °C (V)	1076	296, +05	**Z**

Symbols as in Table 1. Results are for single specimens only. The first row in each site gives the magnetization after AF demagnetization in the peak fields specified. Rows 2 to 4 give the magnetization after thermal demagnetization at the temperatures specified. Rows 5 to 7 give the vector differences between the first row and rows 2 to 4, respectively.

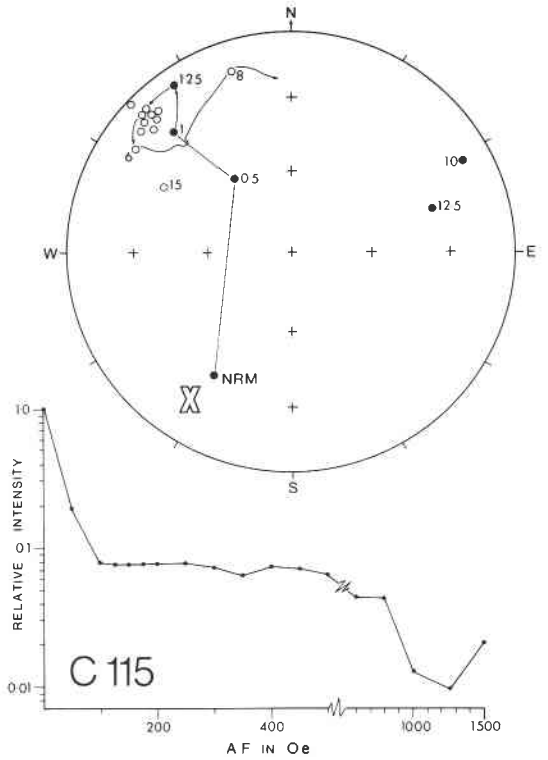


FIG. 5. Alternating field demagnetization of gabbro from site C115. The decay in intensity is shown below. The corresponding direction changes are given above, open (solid) symbols denoting upward (downward) inclinations. AF field strengths are given in hundreds of oersted. Note the apparent stable end-point between 150 and 600 Oe, and the scattered directions and variable intensities produced in higher fields. This apparent end-point is not a single magnetization but a combination of **Y** and **D** magnetizations that can be resolved by thermal demagnetization (Fig. 6). The initial (nrm) intensity is 4045 μG.

direction labelled C in Fig. 4) is that for which scatter among specimens is least. The choice is not critical. The correcting effect of the vector subtraction is small, because **X** is large compared to **D** (Table 2, Fig. 5).

The main opaque phases are ilmenite commonly rimmed with leucoxene, and magnetite with fine and coarse ilmenite lamellae. The magnetite–ilmenite grains are large (100 to 1000 μm), but because of intergrowths the effective grain size of the magnetite is probably much less. The large **X** magnetization presumably resides in this magnetite. Minor opaque phases are hematite, pyrite partly altered to geothite, and pyrrhotite partly altered to marcasite. The hematite occurs dispersed in silicate minerals as very fine needles (1–5 μm) commonly oriented parallel to the cleavage, and it is considered to be the source of the **D** magnetization.

Gabbro, Site C115

The results of AF demagnetization of a single specimen of gabbro at site C115 is shown in Fig. 5. The nrm direction is initially SSW, but the direction moves rapidly to the NW during demagnetization. There is an apparent stable end-point between 150 and 500 Oe, but in higher fields the directions scatter. Apparent stable end-points with similar NW directions occur in all specimens from this site. The corresponding intensity falls rapidly to about 8% of the original value in 100 Oe, remains essentially constant to about 500 Oe, and thereafter proceeds to fall irregularly. Resolution of this apparent end-point magnetization (the direction is labelled AF in Fig. 6) by thermal demagnetization is shown in Fig. 6, in which the average results of all specimens from site C115 are given. Up to 500 °C, the mean direction oscillates in a rather

irregular way around the apparent end-point direction, but at 600 °C the inclination increases, and the direction moves into the field of the **D** magnetization and remains there until 650 °C. Thus the apparent end-point magnetization actually consists of two magnetizations: a **D** magnetization and a magnetization obtained by subtracting vectorially the **D** magnetization from the apparent end-point magnetization. This has a NW declination with negative inclination. It is labelled AF (C) in Fig. 6 and is referred to as the **Y** magnetization. The separation of the **D** and **Y** magnetizations in a single specimen is shown in Table 2. Thus, the nrm consists of three components, **D** and **Y** together with an **X** magnetization (marked NRM (C) in Fig. 6), which is obtained by subtracting **D** and **Y** from the nrm. The mean directions of these three magnetizations are listed in Table 1. The corresponding intensity variations are shown in the lower part of Fig. 6. They are irregular, but there is a drop in intensity between 550 and 600 °C that corresponds to the rotation of the direction into the **D** field. Apparently the blocking temperature of the **Y** magnetization is predominantly between 550 and 600 °C, whereas the blocking temperature of the **D** magnetization is between 600 and 650 °C or above. This suggests that **Y** and **D** are carried by magnetite and hematite, respectively.

The oxide phases are similar to those at site C111 except that the ilmenite is fresh, leucoxene rare, and sulphides are unaltered. The **X** and **Y** magnetizations presumably reside in the magnetite of the magnetite–ilmenite grains, the former in the larger grains. The hematite needles in the silicates are the probable carriers of the **D** magnetization.

Pyroxenite, Site C117

The results of AF demagnetization of a single specimen from site C117 are given in Fig. 7. There is an apparent stable end-point between 100 and 350 Oe, and the magnetization is directed toward the NW. In higher fields the direction almost, but not quite, reverses and then proceeds to scatter. The intensity falls steadily through the apparent stable end-point range to about the 1% of the nrm. No other specimen from this site showed this apparent reversal, and there is no evidence for it from thermal demagnetization. Thermal demagnetization of the apparent stable end-point magnetization is shown in Fig. 8. The directions are little changed up to

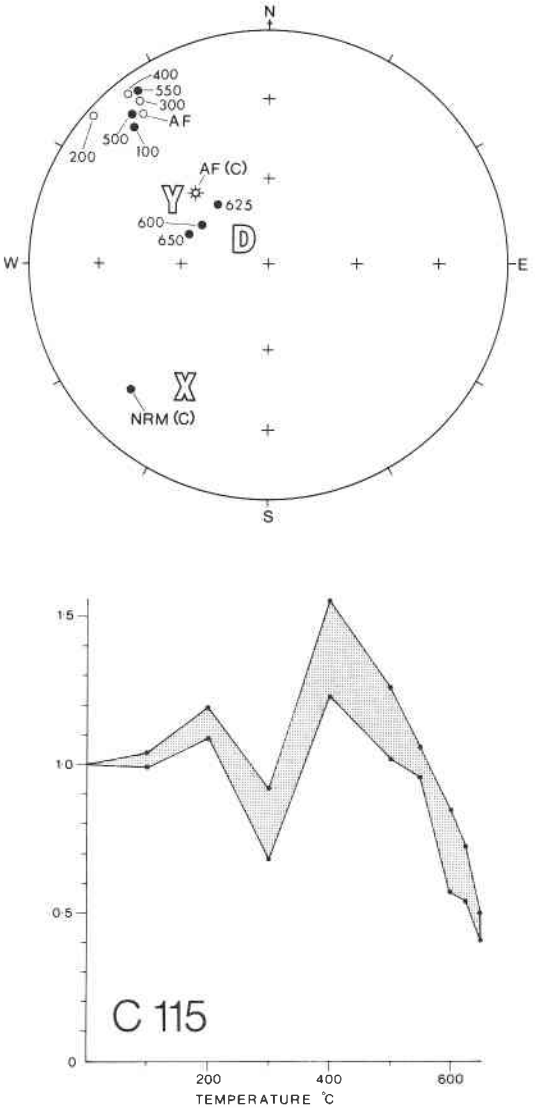


FIG. 6. Thermal demagnetization subsequent to AF demagnetization of gabbro from site C115. Mean directions for 5 core samples are given above. The results have been obtained by thermal demagnetization of specimens that have first been partially demagnetized in 400 Oe. AF is their mean direction after this AF cleaning. The mean directions obtained by subsequent thermal demagnetization are labelled with their corresponding temperatures in °C. Directions with downward (upward) inclinations are marked by dots (open circles). NRM(C) is the nrm corrected for the presence of **Y** and **D** magnetization. AF(C) is the **Y** magnetization direction, with upward inclination, obtained by correcting for the **D** magnetization. The areas of the stereogram occupied by the **X**, **Y**, and **D** magnetizations are indicated. The decay of intensity is shown below by the standard error envelope of 5 thermal demagnetization curves.

500 °C. At 550 °C the inclinations have increased, and at 600 °C the directions enter the **D** field and remain there until 650 °C. The corresponding intensity variations are shown in the lower part of Fig. 8. There is a well marked shoulder above 500 °C, corresponding to the migration into the **D** field. Evidently the apparent stable end-point magnetization consists of two magnetizations, a **D** magnetization and a magnetization obtained by subtracting vectorially **D** from the apparent stable end-point magnetization. This has a NW direction with low inclination (labelled AF(C)) in Fig. 8, and is referred to as the **Z** magnetization. The separation of the **D** and **Z** magnetization in one specimen is given in Table 2. The mean directions of the **D** and **Z** magnetizations at this site are listed in Table 1. The **Z** magnetization has a blocking temperature predominantly between 500 and 550 °C. The **D** magnetization has blocking temperatures between 600 and 650 °C or above. The magnetization removed during demagnetization at 50 Oe is essentially antiparallel to the **X** magnetization

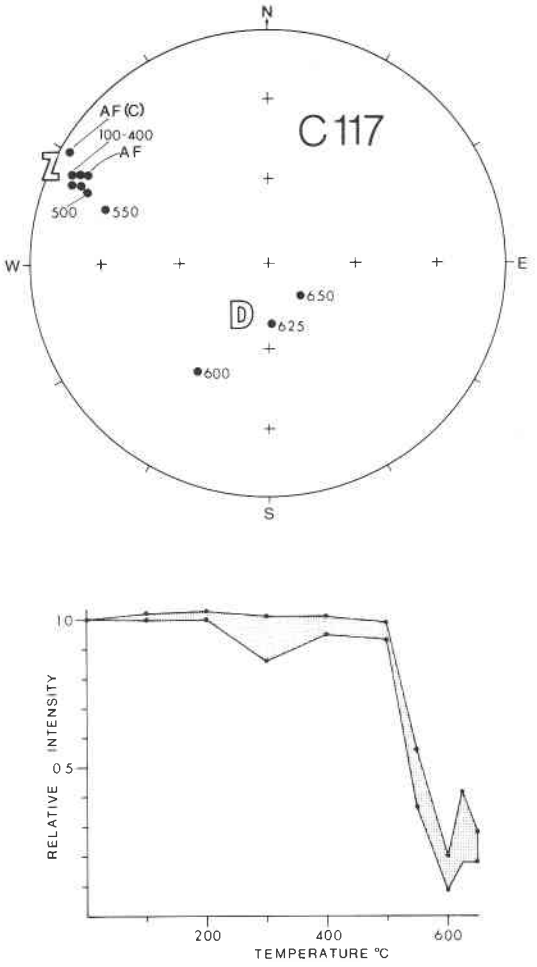


FIG. 8. Thermal demagnetization subsequent to AF demagnetization of pyroxenite from site C117. Mean directions of 4 specimens are given above. All have downward inclinations. Results have been obtained by thermal demagnetization of specimens that have first been partially demagnetized in AF of 200 Oe. The direction labelled AF is the mean direction after demagnetization in AF of 200 Oe. The mean directions obtained by subsequent thermal demagnetization are labelled by their corresponding temperatures in °C. The direction labelled AF(C) is the **Z** magnetization obtained by subtracting the **D** magnetization from AF. The general areas of the stereogram occupied by the **Z** and **D** magnetizations are indicated. The decay of intensity is shown below as the standard error envelope of 4 demagnetization curves.

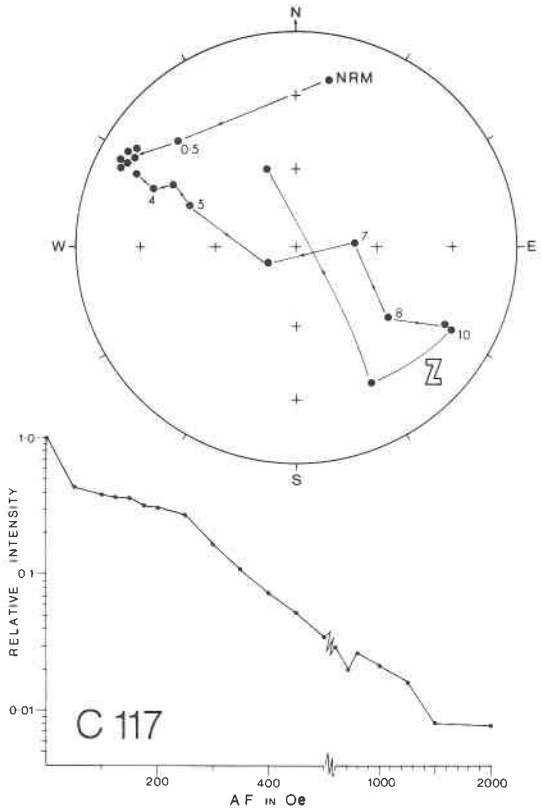


FIG. 7. Alternating field demagnetization of pyroxenite from site C117. The decay in intensity is shown below. The corresponding direction changes are shown above. The AF field intensities being given in hundreds of oersted. All directions have downward inclinations. Note the apparent stable end-point between 100 and 350 Oe. The initial (nrm) intensity is 2483 μG.

of sites C111 and C115. There is, however, no coherence between the removed magnetizations in other specimens at this site and no significance is attached to them.

The oxide phases are similar to those observed at site C111 except that the ilmenite is fresh and leucoxene rare. The magnetite–ilmenite grains contain small blebs of a mineral that could not be firmly identified, but which may be hematite. As before, hematite occurs as fine needles in the silicates, and in addition, larger blebs of magnetite and hematite also occur in the silicates. Hematite is presumably the carrier of the **D** magnetization. The **Z** magnetization presumably is carried by magnetite.

Syenite

The magnetization of the syenites is much simpler. **D** magnetizations predominate. **Y** magnetizations only are found at two sites (A115, B20). The AF demagnetization curves again fall rapidly (Fig. 9), and usually move into the **D** field after demagnetization in fields between 100 and 300 Oe. Thermal demagnetization after AF cleaning (the intensity is then usually between 5 and 10% of nrm) yields the mean curve of Fig. 10 with blocking temperatures between 500 and 600 °C, indicating that magnetite is responsible for the **D** magnetization.

The dominant opaque mineral is ilmenite, occasionally extensively altered to leucoxene. Magnetite occurs as small discrete grains associated with ilmenite and as inclusions in ilmenite. Although magnetite is apparently responsible for remanence, it is not seen in many sections, but only very small quantities are required to produce the observed remanence. Amorphous hematite occurs as fine dust in some of the silicates.

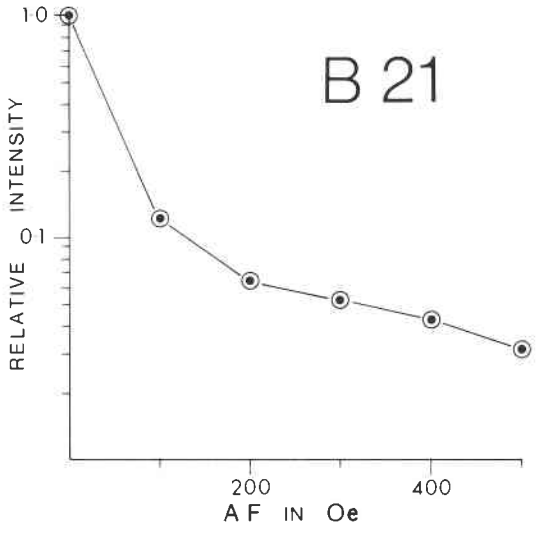


FIG. 9. Alternating field demagnetization of syenite from site B21. The nrm intensity is 983 μG.

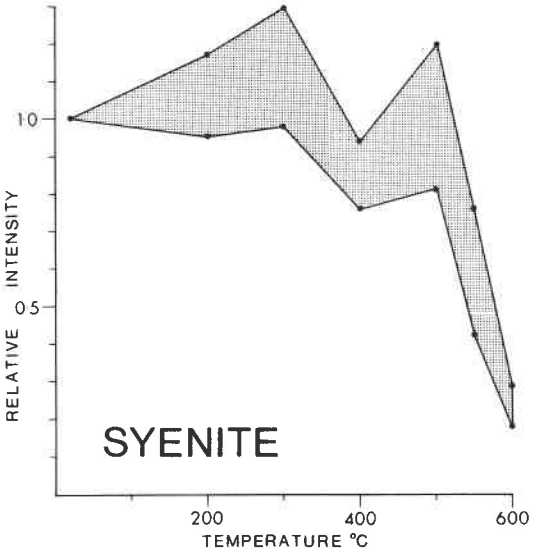


FIG. 10. Thermal demagnetization of syenite. The standard error envelope is shown of 10 demagnetization curves obtained by thermal demagnetization of syenite specimens after AF demagnetization in 100 to 200 Oe is shown.

Analysis of Directions

Of the 22 sites sampled, all but two gave coherent data. At sites A114 and C112, scattered directions were observed and no coherent magnetization could be found. The magnetizations at each collecting site are listed in Table 1 and plotted in Fig. 11. The mean directions of the four magnetizations are given in Table 3. All syenite sites except B20 and A115 yielded **D** magnetizations. B20 and A115 have **Y** magnetizations only. Most sites in basic rocks from the northern part of the complex yielded several magnetizations, and the **D** magnetization is present at all except site C118. The **X**, **Y**, and **Z** magnetizations are distributed in an irregular way, and there is no systematic relationship to rock types. The **D** magnetization is present at most sites (19 out of 20). The **Y** magnetization contains reversals. The separation of the **X**, **Y**, and **Z** magnetizations is arbitrary, and although there do seem to be three distinct groups, it could

TABLE 3. Summary of results

Magnetization	$B(N)$	D,I	k	$\delta°$	$\alpha°_{95}$	Pole	$A°_{95}$	$\%N$
D	19(85)	022,+88	17	19	8	67° N 113° W	15	100
Y	5(25)	137,+46	67	12	9	07° N 078° W	10	60
Z	4(16)	291,+02	14	19	26	10° S 044° W	16	0
X	2(9)	210,+27	—	—	—	09° S 145° W	—	100

B is the number of collecting localities and N is the number of cores. Since more than one magnetization is present in the same specimen, the sum of the B's (= 30) is greater than the number of sites (= 20; two, A114 and C112, gave no useful results), and the sum of the N's (= 135) is greater than the number of cores used (= 101). Unit weight given to each locality in the calculations. D,I are the declination and inclination of the mean direction; k and α_{95} are Fisher's precision and circle of confidence ($P = 0.05$) (Fisher 1953) of the directions; $\delta°$ is Wilson's angular standard deviation (Wilson 1959). $A°_{95}$ is the circle of confidence of the corresponding poles. $\%N$ is the percentage of normal polarities, assuming the poles of Figs. 13 and 14 are north poles.

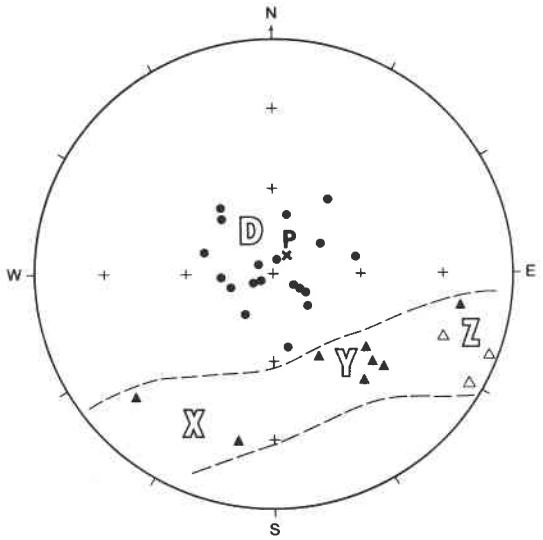


FIG. 11. Summary of directions of magnetizations in the Big Spruce Complex. Solid (open) symbols denote downward (upward) inclinations. The **D** magnetization is marked by circular dots. Two of the **Y** and all the **Z** magnetizations are reversed, and they are plotted here in the SE quadrant irrespective of sign, for ease of comparison. The **X**, **Y**, and **Z** magnetizations are denoted by triangles. P is the present geomagnetic field direction. The broken line outlines the zone containing the **X**, **Y**, and **Z** magnetizations.

also be argued that they form a discontinuous belt of directions extending from E to SW (Fig. 11).

In Figs. 12 and 13, the poles corresponding to the magnetizations observed in the Big Spruce Complex are plotted and compared with paths of apparent polar wander (APW) constructed previously. In Fig. 12, the pole corresponding to the **D** magnetization and other poles for the interval 2200 to 1800 m.y. are shown. They form a reasonably coherent array, stretching from what is nowadays the Mackenzie delta to Peru. This APW path is referred to as track 5. Readers are referred to papers by McGlynn *et al.* (1975) and Irving and McGlynn (1976) for a discussion of track 5 and its tectonic significance. One notable feature is the excellent agreement between the poles for the Otto Stock and the **D** magnetization of the Big Spruce Complex. These bodies have statistically similar Rb–Sr ages (see caption to Fig. 12). Studies of the metamorphic aureole of the Otto Stock have shown that the remanence was acquired during cooling following instrusion (Pullaiah and Irving 1975). Moreover, the **D** magnetization has directions that are different from those in any neighbouring rock unit. Hence, it is reasonable to suppose that the **D** magnetization of the Big Spruce Complex was acquired at the time of initial cooling, between 2150 and 2070 m.y.

The mean direction of the **D** magnetization is not significantly different from the present field (Fig. 11 and Table 3) and it might be argued, therefore, that it is a very tenacious viscous or weathering magnetization of recent origin, aligned along the present field. It seems to us that this is very unlikely, because the blocking temperatures are very high; because the carrier of the **D** magnetization in the syenites is magnetite, not hematite (Fig. 10); because the rocks are fresh, being taken from cores drilled from beneath the outer crust; and because there is no evidence of weathering of magnetite or hematite in polished and thin sections. The crystalline hematite in the silicates of the basic rocks is apparently a product of primary crystallization.

In Fig. 13, a reconstruction of the APW path for the interval 1800 to 1300 m.y. is shown. This is referred to as track 4. References may be made to Park *et al.* (1973), McGlynn *et al.* (1974), and Irving and McGlynn (1976) for discussions of track 4. Tracks 4 and 5 cross one another in

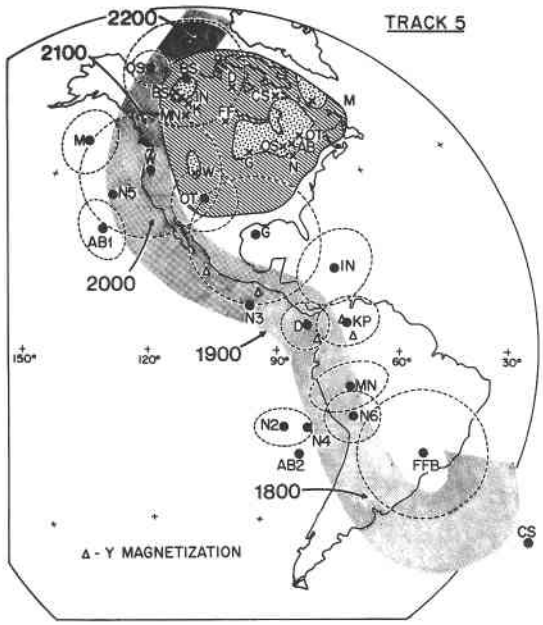


FIG. 12. Track 5. The reconstructed polar path is represented by a broad band calibrated every 100 m.y., and is taken from McGlynn *et al.* (1975). Error circles at $P = 0.05$ are given. The poles and collecting localities are indexed as follows: AB1 and AB2 — Abitibi dikes (Larochelle 1966), Rb–Sr isochron 2147 ± 68 m.y. (Gates and Hurley 1973); D—Dubawnt Group, Rb–Sr isochron 1830 ± 26 m.y. (Park *et al.* 1973); G—Gunflint Formation, ≃2000 m.y. (Symons 1966); IN—Indin dikes (McGlynn and Irving 1975), Rb–Sr isochron 2093 ± 86 m.y. (Gates and Hurley 1973); KP—Kahochella Formation, Rb–Sr isochron from associated volcanics 1873 ± 13 m.y. (McMurry *et al.* 1973); M—Mugford basalt (Murthy and Deutsch 1972), preliminary Rb–Sr isochron ≃2300 m.y. (Barton 1973); MN—Martin Formation, K–Ar on associated dikes 1835 m.y. (Evans and Bingham 1973); N—Nipissing Diabase, Rb–Sr isochrons 2160 ± 60 and 2150 ± 50 (Fairbairn *et al.* 1969; Van Schmus 1965), five pole determinations—N2 (Symons 1970), N3 (Symons 1971), N4 (selected sites from Symons 1970), N5 (Symons 1970), N6 (Patel and Palmer 1974); OT—Otish Gabbro, age from indirect evidence ≃2000 m.y. (Fahrig and Chown 1973); OS—Otto Stock original magnetization (Pullaiah and Irving 1975), Rb–Sr isochron 2140 ± 80 m.y. K. Bell and J. Blenkinsop manuscript submitted to Can. J. Earth Sci., June 1975; W—Wind River dikes (Spall 1971), maximum K–Ar 2060 m.y.; BS—Big Spruce Complex, **D** magnetization (herein); FFB is the **B** magnetization observed in the Amisk and Boundary Intrusions at Flin Flon and considered to be about 1800 m.y. old by Park (1975). The poles for the **Y** magnetization are indicated by open triangles.

what is now Central America. Using this track as a scale for determining the age of magnetization, then the **Z** magnetizations (poles near eastern South America) appear to be *about* 1800 m.y. old, and the **X** magnetizations (poles in the central Pacific) appear to be *about* 1400 m.y. old. The poles for the **Y** magnetizations (poles near Central America), however, fall on the cross-over of tracks 4 and 5, and hence may be assigned an age of either *about* 1900 or *about* 1600 m.y. It is noteworthy that the **Y** magnetizations are similar in direction to those observed from the Indin Dikes (2093 ± 86 m.y., Gates and Hurley 1973), which approach to within 20 km of the Big Spruce Complex (Fig. 1). Also, Martineau and Lambert (1974) have obtained and Rb–Sr whole-rock isochron of 1926 ± 40 m.y. from a nearby biotite pyroxenite complex. Thus there is evidence of thermal events near the Big Spruce Complex between 2100 and 1900 m.y. BP, and the **Y** magnetizations could have been acquired during that interval. The **Z** magnetization may correspond in time to the K–Ar biotite age of 1785 m.y. from the Big Spruce Complex (Leech *et al.* 1963), which presumably reflects post-Hudsonian uplift. Thus, the **Y** and **Z** magnetizations may be associated with identifiable local thermal events at *about* 2000 and 1800 m.y. BP, respectively. There is no known local thermal event about 1400 m.y. ago with which the **X** magnetization could be identified.

Origin of Polyphase Magnetization

In the previous section approximate ages spanning about 800 m.y. have been assigned to the four magnetizations. A number of blocking temperature ranges are present, which indicates that the magnetizations are carried by different iron oxide phases. Observation of polished sections shows that several oxide phases are indeed present. The polyphase magnetization therefore could have originated either by changes in the composition or grain-size relationships of the iron oxides, or by thermal processes operating over this 800 m.y. interval. The geochronological evidence indicates a complex thermal history. There could have been a single episode of very slow cooling, or a number of heating and cooling episodes. Thermal and chemical processes may have operated together. The magnetite observed in polished sections is fresh, and the hematite grains within the silicates are also fresh, and there is no reason to believe that they are not the product of initial crystallization. Alteration of ilmenite to leucoxene and sulphides to goethite occur at 5 sites (B20, B22, B24, B29, C111), but

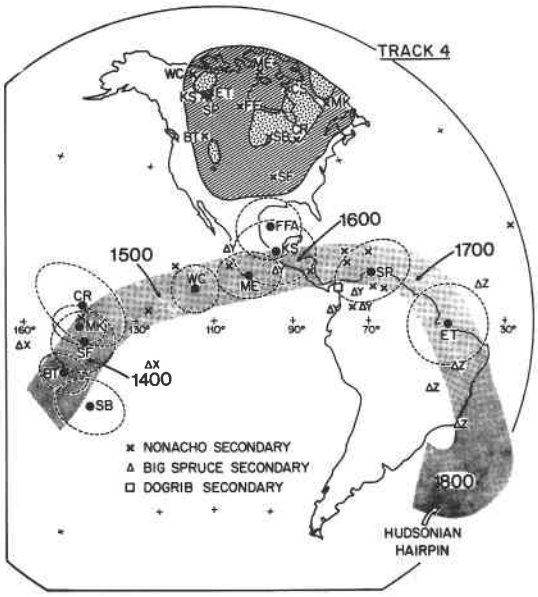


FIG. 13. Track 4. Reconstructed polar path is represented by a broad band calibrated every 100 m.y., and is taken from Irving and McGlynn (1976). Pole positions for the interval 1800 to 1400 m.y. are plotted and their errors given: ET—Et-then Group redbeds (Irving *et al.* 1972); SP—Sparrow Dikes (McGlynn *et al.* 1974) preferred K–Ar age 1700 m.y.; FFA—Flin-Flon metaigeneous and sedimentary rocks, the younger A component considered by Park (1975) to correspond in time to K–Ar ages of about 1700 m.y.; KS—Kahochella redbeds, secondary magnetization (McMurray *et al.* 1973); ME—Melville–Daly Bay basic metamorphic rocks (Park 1973), K–Ar age 1650 m.y.; WC—Western Channel diabase (Irving *et al.* 1972), K–Ar age of 1500 m.y.; MK—Michikamau anorthosite (Murthy *et al.* 1968), K–Ar age 1400 m.y.; SF—St. Francois igneous rocks (Hsu *et al.* 1966), Rb–Sr isochron 1400 m.y.; SB—Sibley Series redbeds (Robertson 1973), Rb–Sr isochron 1370 ± 32 m.y.; BT—Belt Series redbeds (Collinson and Runcorn 1960); CR—Croker Island Complex (Palmer 1969), Rb–Sr isochron 1475 m.y. Overprints attributed to thermal events occurring long after initial formation are plotted as open triangles: Nonacho Group sediments (McGlynn *et al.* 1974); **X**, **Y**, and **Z** magnetization of Big Spruce Complex (herein); Dogrib Dikes (McGlynn and Irving 1975).

it is impossible to say whether this alteration is deuteric, or whether it occurred long after initial cooling.

If the age assignments are correct, and if there has been no changes in the composition and grain-size relationship of the iron oxides, then the older magnetizations ought to have the higher blocking temperatures. The blocking temperatures of the **D** magnetization range from 500 to 650 °C, the range in about one half of the specimens being 600 to 650 °C. The blocking temperatures of the **Y** magnetization are mainly between 500 and 600 °C, at one site (C118) between 600 and 650 °C. The blocking temperatures of the **Z** magnetization are in the range 500 to 600 °C. The blocking temperatures of the **X** magnetization are between 200 to 400 °C. The highest blocking temperatures (above 600 °C) occur in the **D** magnetization associated with hematite in the basic phases of the northern part of the intrusion. In the **Z** and **X** magnetizations, such high blocking temperatures are not found, and crystalline hematite is not observed. In the **Y** magnetization, blocking temperatures as high as 600 °C occur only at one site. Thus, there is a general increase of blocking temperature with apparent age. But the correlation is imperfect, different magnetizations sometimes having similar ranges of blocking temperatures; for example, the **D** magnetization in the syenites has the same blocking temperature range as the **Z** magnetization in the basic phases (compare Figs. 8 and 10). Some of this apparent overlap in the blocking temperature spectra of different magnetizations could arise from experimental error because of noise in the two stage demagnetization procedure employed, but the high blocking temperature of the **Y** magnetization at C118 is real. Hence, although the hypothesis that the polyphase magnetization is caused by thermal effects without changes in the iron oxides explains the results in a general way, it does not account for the fact that the blocking temperature ranges of some of the magnetizations overlap. As a first approximation, we suggest that cooling following a succession of mild heating episodes and final post-Hudsonian uplift in the time range 2000 to 1400 m.y. is the prime cause of the observed overprints, and that imperfections in the expected correlation between blocking temperature and apparent age arise from experimental uncertainties, and from minor changes at some sites in the composition and the grain-size relationships of the iron oxides. Although the study of magnetic overprints is in its infancy, it is already clear that it is capable of providing important information about the detailed geological history of rocks.

Since compiling Fig. 11, Symons and Londry (1975) have given two further poles from the Nipissing diabase, at 14° S 100° W and at 67° N 111° W, both of which fall on track 5. The Nipissing pole at 67° N 111° W agrees very well with

the poles from the Otto Stock and the Big Spruce Complex, and the three bodies are, within the errors, of the same age. Symons and Londry, however, reject the pole at 67° N 111° W as being due to magnetization by the present field; but the magnetizations are just as stable as other magnetizations from the Nipissing diabase (Symons and Londry 1975, their fig. 4), and Pullaiah and Irving (1975) have shown that similarly directed magnetizations in the Otto Stock are not caused by the present earth's field, but were acquired during cooling after emplacement. Hence it is possible, as Irving and McGlynn (1976) have contended, that the second Nipissing pole at 14° S 100° W is an overprint. On this view the Otto, Big Spruce, and Nipissing bodies have the same paleomagnetic pole near the present western Arctic coast of Canada, and they are part of a major rifting event at about 2100 m.y. that followed the deposition of Huronian sedimentary rocks.

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